Harmonic Suppression and Voltage Tracking in Single-Phase Stand-Alone Inverters Using Harmonic Control Arrays

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***Abstract*—In single-phase standalone inverters, harmonic currents drawn particularly by nonlinear loads cause distortions in the output voltage, and this situation adversely affects the proper operation of the connected loads. The harmonic control array (HCA) method disperses the error signal into its harmonic components and controls each harmonic with independent gains, thereby achieving the tracking of periodic references and the suppression of periodic disturbances simultaneously. In this article, the HCA method is applied to control the output voltage of a single-phase full-bridge standalone inverter with an inductor-capacitor (LC) filter under linear and nonlinear load conditions. A single set of pre-tuned gains controls selected harmonics under both linear and nonlinear load conditions without adaptation. The performance of the method has been validated through simulation and experimental studies. Under both load conditions, the HCA method successfully reduces output voltage harmonics while maintaining precise tracking of the reference signal.**

***Index Terms*—Harmonic Control Array (HCA), harmonic suppression, nonlinear load, periodic reference tracking, single-phase stand-alone inverter, total harmonic distortion (THD).**

I. Introduction

SINGLE-PHASE voltage source inverters are the most widely used power electronic converters in standalone applications, such as uninterruptible power supplies (UPSs) and distributed generation units [1]. In these applications, the primary function of the inverter is to provide a sinusoidal output voltage with regulated amplitude and frequency, and low total harmonic distortion (THD), regardless of load variations [1], [2]. For UPSs, the output voltage performance and test conditions under linear and nonlinear loads are defined in the IEC 62040-3 standard [3]. Harmonic currents drawn by nonlinear loads, such as diode rectifiers, create harmonic voltage drops across the output filter and inverter impedance, thereby distorting the waveform of the output voltage [4], [5]. Although the inductor-capacitor (LC) output filter suppresses components at the switching frequency, it is not sufficient on its own to mitigate low-order load harmonics [2]. Therefore, the low-order harmonics in the output voltage under nonlinear loads must be suppressed by the controller.

Numerous methods have been proposed in the literature for the output voltage control of standalone inverters. Multi-loop structures, consisting of an outer voltage loop and an inner current loop, are widely proposed because they provide active damping and good transient performance [6], [7]. For single-phase systems, approaches such as voltage control in the synchronous reference frame (SRF) [6] and single-loop dq control [2], which stands out for its simple implementation and fast response, have also been developed. In addition to these, Lyapunov-based adaptive control [8], model predictive control [9], and a control scheme that does not require an output voltage sensor [1] have also been applied to standalone inverters. Furthermore, a comparative study of discrete-time proportional-integral (PI) and proportional-resonant (PR) controllers for single-phase UPS inverters is presented in [10]. Recent studies have also investigated precisely shaping the output impedance of the inverter at selected harmonic frequencies [11], while others have achieved both clean voltage in standalone mode and clean current in grid-connected mode using a single control structure [12]. While each of these methods offers a different trade-off between dynamic performance, robustness, and implementation complexity, the suppression of specific harmonics under nonlinear loads with zero steady-state error generally requires additional harmonic compensation structures [2], [6].

The two most common approaches for tracking periodic references and suppressing periodic disturbances are resonance-based controllers and repetitive controllers. The PR controller can track a sinusoidal reference in a stationary frame with zero steady-state error by providing high gain at the resonant frequency [13], [14]. For harmonic suppression, multi-resonant

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structures are obtained by adding a separate resonant term for each harmonic frequency. These structures have been successfully employed in UPS inverters [15] and standalone or weak-grid-connected converters [16], [17]. Controlling a higher number of harmonics expands the total tuning parameters, requiring a comprehensive stability analysis for each individual resonant term within the closed loop [16].

Repetitive control is based on the internal model principle [18] and provides high gain simultaneously at all harmonics of the fundamental frequency by introducing a fundamental-period delay into the loop [19]. Thanks to this feature, it is effectively used in the compensation of unbalance and harmonic distortion in UPS inverters [20], standalone distributed generation inverters [21], and voltage source inverters supplying nonlinear loads [22]. A comprehensive review of repetitive control applications in power electronics is provided in [23]. In this approach, stability and robustness design, along with sensitivity to fundamental frequency variations, emerge as key design considerations.

Unlike these methods, in some studies, the error or the measured signal is first dispersed into its harmonic components, and then each harmonic is compensated individually. For instance, in [24], the amplitude and phase of each harmonic are accurately detected using a digital lock-in amplifier (DLA), and the reconstructed harmonic is compensated via negative feedback.

The harmonic control arrays (HCA) method is also based on this concept, but it incorporates harmonic dispersion directly into the control structure [25]. By isolating and controlling the error signal's harmonic components in the complex domain before recombining them, the HCA method enables individual compensation of desired frequencies and DC terms. Bounded only by processor capacity, this architecture effectively avoids the intensive coordinate transformations usually associated with multi-harmonic vector control [25].

The HCA method was introduced in [25] with a real-time application for periodic position control and was subsequently applied to single-phase standalone inverters [26] and active power filters [27]. In recent studies, the method has been extended to various problems, such as periodic position tracking in DC motors [28], control of chaotic systems [29], and multi-harmonic periodic disturbance compensation in systems with large input delays [30].

In [26], where the HCA method is applied to a standalone inverter, the real-valued PI gains for the fundamental component are designed based on a simplified model of the inverter, while the gains for higher harmonics are obtained by dividing the fundamental component gains by the harmonic order. Under the nonlinear load defined in the IEC 62040-3 [3] standard, the total harmonic distortion (THD) of the output voltage is reduced from 4.2% to 2.6% by compensating the 3rd and 5th harmonics in addition to the fundamental component, and it is noted in [26] that more harmonics can be suppressed at the expense of a higher computational burden. Complex-valued gains, on the other hand, allow for the independent adjustment of the amplitude and phase of each harmonic, though they require the selection of two complex numbers for each harmonic [25]. Although a tuning rule for complex PI gains in systems with time delays is proposed in [30], how to tune a complex-gained HCA for multiple harmonics in an LC-filtered inverter and implement it in real-time on a low-cost processor has not yet been investigated.

In this article, the HCA method with complex-valued gains is applied to the output voltage control of an LC-filtered single-phase full-bridge standalone inverter. In the proposed structure, the harmonics to be controlled are selected based on load conditions, and each harmonic is controlled independently using its own complex gain. The performance of the method is evaluated through simulation and experimental studies under linear and nonlinear load conditions. The main contributions of the study are as follows:

1) The HCA method with complex-valued gains is applied to the output voltage control of a single-phase standalone inverter. Thus, higher-order harmonics are selectively compensated alongside the fundamental component by independently adjusting the amplitude and phase of each control signal harmonic.
2) To determine the complex gains, a systematic tuning procedure is presented, in which the phase and amplitude of the integral gain for each harmonic are tuned sequentially, progressing from lower-order harmonics to higher-order harmonics.
3) The harmonics to be controlled are selected based on the load conditions, and the same set of gains is utilized under both linear and nonlinear load conditions without requiring gain retuning.
4) The HCA controller is implemented in real time on a low-cost microcontroller, and the effectiveness of the method is validated through simulation and experimental results.

The remainder of the article is organized as follows. Inverter and load models are described in Section II, and the HCA method is explained in Section III. Section IV presents the discrete-time implementation and the selection of complex gains, Section V provides simulation and experimental results, and Section VI concludes the study.

II. Single-Phase Full-Bridge Inverter

The system configuration investigated in this study consists of a single-phase full-bridge inverter connected to an LC output filter, as illustrated in Fig. 1. To mitigate ground-loop issues and ensure electrical safety, the inverter is supplied from a variable DC autotransformer, which provides an adjustable DC output in the range of 0–450 V and is isolated from the 380 V three-phase utility grid via a 3 kVA isolation transformer. In this study, the autotransformer output was set to provide a DC bus voltage of $V_{dc} = 200$ V.

The switches of the bridge are driven by unipolar sinusoidal pulse-width modulation (SPWM) signals generated by the microcontroller, and the gate signals are transmitted to the inverter board through an optocoupler-based isolation circuit. Denoting the filter inductance and capacitance by L_f and C_f, and their equivalent series resistances by r_L and r_C, the output stage is described by

$$L_f \frac{\mathrm{d}i_L}{\mathrm{d}t} = v_i - r_L\, i_L - v_o, \tag{1}$$

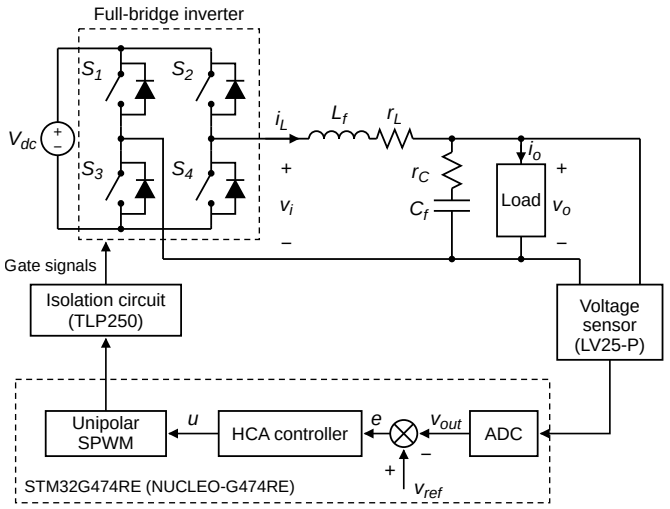


Fig. 1. Single-phase full-bridge inverter with LC output filter, voltage sensor, isolation circuit and microcontroller. ADC: analog-to-digital converter.

TABLE I
SYSTEM PARAMETERS FOR SIMULATIONS AND EXPERIMENTS

Parameter	Symbol	Value
Power Circuit		
DC Bus Voltage	V_{dc}	200 V
Reference Voltage	v_{ref}	167.5 V_{peak}
Filter Inductance	L_f	3 mH
Filter Capacitance	C_f	21 µF
Inductor Series Resistance	r_L	0.6 Ω
Capacitor Series Resistance	r_C	3.5 Ω
Switching Frequency	f_{sw}	20 kHz
Sampling Frequency	f_s	40 kHz
Fundamental Frequency	f	50 Hz
Dead time	t_d	400 ns
Modulation Index	m_a	0.85
Microcontroller Unit (MCU)	-	STM32G474RE
Linear Load		
Load Inductance	L_L	0.7 mH
Load Resistance	R_L	20 Ω
Nonlinear Load		
Series Input Resistance	R_s	1 Ω/ 200 W
DC-Side Capacitance	C_{dc}	1000 µF / 450 V
Balancing Resistors	R_b	2 × 22 kΩ
Resistive Load Steps	R_{load}	125 Ω

$$C_f \frac{\mathrm{d}v_C}{\mathrm{d}t} = i_L - i_o, \qquad v_o = v_C + r_C\,(i_L - i_o)\,, \tag{2}$$

where v_i is the bridge output voltage, i_L the inductor current, v_C the capacitor voltage, v_o the load voltage and i_o the load current. Averaged over one switching period, the bridge output voltage can be modeled as $v_i \approx u\,V_{dc}$, where u denotes the modulating signal. In the prototype, $r_L = 0.6\ \Omega$ and $r_C = 3.5\ \Omega$. Based on the filter parameters in Table I, the undamped resonance frequency of the filter, $1/(2\pi\sqrt{L_f C_f})$, is approximately 634 Hz. Positioning this resonance peak between the 11th and 13th harmonics ensures that the physical filter network does not inadvertently amplify dominant low-order distortion frequencies. To complement this physical layout, the HCA framework is selectively deployed up to the 21st harmonic under the nonlinear load, enabling precise active compensation across the filter's resonance envelope.

III. HARMONIC CONTROL ARRAYS METHOD

The HCA method was developed for control systems with periodic references or disturbances [25], [31]. Instead of processing the real-time error signal directly, as in classical controllers, the HCA method separates the error into a predetermined number of complex harmonic components, applies a separate controller to each component, and recombines the resulting control components into a real-valued control signal. The method consists of three blocks, shown in Fig. 2: the harmonic disperser, the controller layer and the harmonic assembler. The formulation below follows [31].

Harmonic disperser: Let T be the fundamental period, $\omega = 2\pi/T$ the fundamental angular frequency, and H the highest harmonic number considered. The disperser obtains the hth harmonic of the error signal $e(t) = v_{ref}(t) - v_{out}(t)$ by a running Fourier series integral over the last fundamental period,

$$\langle e\rangle_h(t) = \frac{1}{T}\int_{t-T}^{t} e(\tau)\,\exp(-jh\omega\tau)\,\mathrm{d}\tau. \tag{3}$$

$\langle e\rangle_h$ is a complex-valued, time-varying Fourier coefficient. If $e(t)$ is periodic with period T, $\langle e\rangle_h$ becomes a complex constant equal to the hth Fourier series coefficient. Since $e(t)$ is real valued, negative harmonics need not be calculated, and the harmonics are collected in the vector $\langle e\rangle = [\langle e\rangle_0, \langle e\rangle_1, \ldots, \langle e\rangle_H]^T$.

Controller layer: Because the harmonic components of the error become complex constants in periodic steady state, an integral action on each component is needed to drive its steady-state error to zero, while a proportional action can be added to shorten the transient. In this work, the controller layer is built from PI controllers,

$$\langle u\rangle(t) = K_P\langle e\rangle(t) + K_I\int_0^t \langle e\rangle(\tau)\,\mathrm{d}\tau, \tag{4}$$

where $\langle u\rangle = [\langle u\rangle_0, \langle u\rangle_1, \ldots, \langle u\rangle_H]^T$ is the vector of the control harmonics, and K_P and K_I are complex-valued proportional and integral gain matrices. When they are chosen as diagonal matrices, each error harmonic modifies only the corresponding control harmonic, so that every harmonic is controlled by its own PI controller with complex gains. The complex gains allow both the amplitude and the phase of each harmonic of the control signal to be adjusted.

Harmonic assembler: The assembler recombines the harmonic components of the control signal into the real-valued control signal using the Fourier series synthesis,

$$u(t) = \langle u\rangle_0(t) + 2\,\Re\left\{\sum_{h=1}^{H}\langle u\rangle_h(t)\,\exp(jh\omega t)\right\}, \tag{5}$$

which is applied to the modulator as the modulating signal u.

IV. CONTROLLER DESIGN AND IMPLEMENTATION

A. Discrete-Time Implementation

For implementation on the microcontroller, the HCA blocks are realized in discrete time [31]–[33]. With a sampling period

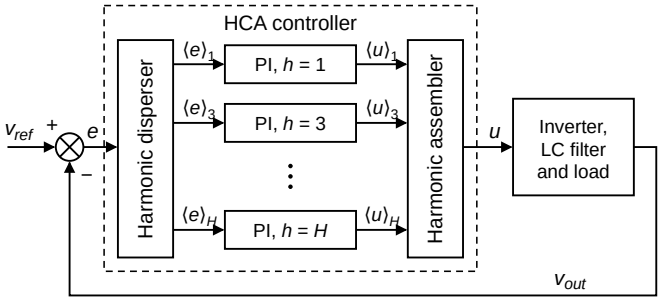


Fig. 2. Block diagram of the HCA-controlled inverter, showing the harmonic disperser, the controller layer of PI controllers and the harmonic assembler.

T_s chosen so that $N = T/T_s$ is an integer, the disperser in (3) becomes

$$\langle e\rangle_h[n] = \frac{1}{N}\sum_{k=n-N+1}^{n} e[k]\exp(-j2\pi hk/N)\,. \tag{6}$$

Evaluating (6) at every sample requires N complex multiplications per harmonic, which is heavy for large N. The equivalent recursive form

$$\begin{aligned}\langle e\rangle_h[n] &= \langle e\rangle_h[n-1] \\ &\quad + \frac{1}{N}\left(e[n] - e[n-N]\right)\exp(-j2\pi hn/N)\end{aligned} \tag{7}$$

is used instead. This recursive update is a form of the sliding discrete Fourier transform (DFT) [34]. Once the exponential terms are precalculated and stored in a table, (7) needs only one complex multiplication per harmonic and a buffer of N real numbers holding the last period of the error signal. The PI controller in (4) and the assembler in (5) are implemented as

$$\langle u\rangle[n] = K_P\langle e\rangle[n] + K_I T_s \sum_{k=0}^{n}\langle e\rangle[k], \tag{8}$$

$$u[n] = \langle u\rangle_0[n] + 2\,\Re\left\{\sum_{h=1}^{H}\langle u\rangle_h[n]\,\exp(j2\pi hn/N)\right\}. \tag{9}$$

In this work, the sampling frequency is 40 kHz and the fundamental frequency is 50 Hz, so that $N = 800$ samples are taken in each fundamental period. Along with the fundamental component, odd harmonics up to $H = 7$ are controlled under the linear load and up to $H = 21$ under the nonlinear load.

B. Selection of the Complex Gains

In the earlier application of the HCA method to a single-phase stand-alone inverter [26], real-valued PI gains were designed for the fundamental component from a simplified model of the inverter, and the gains of the higher harmonics were obtained by dividing the fundamental gains by the harmonic number h. In this work, the gains are complex valued and are set separately for each harmonic. The gain matrices K_P and K_I are diagonal, so each harmonic h has its own pair of complex gains, $K_{P,h}$ and $K_{I,h}$, and both the amplitude and the phase of each harmonic of the control signal can be adjusted.

TABLE II
COMPLEX HCA GAINS ($K_{P,h} = 0$ FOR ALL h)

Harmonic h	$\lvert K_{I,h}\rvert$	$\angle K_{I,h}$ (deg)
1	0.7126	22.51
3	0.2719	27.91
5	0.5946	10.42
7	0.4912	0.63
9	0.1889	51.75
11	0.0471	115.94
13	0.1115	61.78
15	0.0740	140.75
17	0.0224	90.00
19	0.0224	90.00
21	0.5000	90.00

The complex gains were adjusted manually and step by step. In the first stage, only the fundamental component was controlled: the proportional and integral gains associated with all other harmonics were set to zero, and the proportional gain $K_{P,1}$ of the fundamental component was also kept at zero. The integral gain $K_{I,1}$ was then determined in two steps, by first adjusting its phase angle and then its magnitude so as to minimize the total harmonic distortion (THD) of the output voltage. If deemed necessary, once $K_{I,1}$ has been determined, $K_{P,1}$ can also be adjusted according to the same criterion.

After the gains of the fundamental component were determined, the same procedure was repeated for the 3rd and 5th harmonics, respectively. For each harmonic, the phase and magnitude of $K_{I,h}$ were adjusted to mitigate the effect of that harmonic at the output and reduce the THD. In this way, the gains of all controlled harmonics were determined sequentially, starting from the lower harmonics and proceeding to the higher ones, as suggested in [25].

The gain values obtained with this procedure are listed in Table II. The same gain set was used in all simulation and experimental tests. For the linear load, the gains up to the 7th harmonic were applied, and for the nonlinear load, the complete set up to the 21st harmonic was used.

V. PERFORMANCE EVALUATION

The performance of the proposed HCA method has been investigated on a single-phase full-bridge inverter system under linear and nonlinear loads, in both a simulation environment and a real-time experimental setup. The performance of the method was evaluated based on two quantitative metrics: the total harmonic distortion (THD) of the output voltage and the root-mean-square (RMS) tracking error (E_{rms}) between the output voltage and the reference signal. The simulation model and the experimental setup were constructed using the same circuit configuration and the system parameters given in Table I. Since the same controller configuration given in Section IV was used in both environments, the results obtained from the simulation and real-time environments can be compared directly.

The linear load consists of an inductance L_L and a resistance R_L connected in series. The nonlinear load consists of a full-bridge rectifier with a series resistor R_s at its input. An electrolytic capacitor C_{dc}, two balancing resistors R_b, and a five-stage resistive load unit R_{load} are connected in parallel to

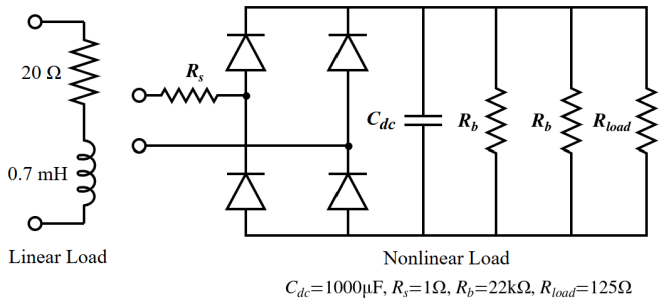


Fig. 3. Configuration of linear and nonlinear loads used in simulation and experimental studies.

the rectifier output. In the simulation and experimental studies, the load unit was set to 125 Ω. Details regarding the structure of the linear and nonlinear loads used in the tests are provided in Fig. 3, while the load parameters are listed in Table I.

In all studies, the modulation index was set to 0.85, and the peak value of the reference voltage was determined to be 167.5 V accordingly. This value was selected to prevent the modulation signal from saturating.

A. Simulation Studies

The system was verified via simulation in MATLAB/Simulink, where dead-time and switch conduction drops were not modeled. The THD calculation includes components up to the 50th harmonic (2500 Hz). Interestingly, the obtained experimental and simulation results are highly aligned. This alignment highlights the disturbance-rejection capability of the HCA framework. Because the HCA loop actively compensates for real-world hardware non-idealities (such as dead-time distortion) in real-time, it effectively matches the physical system's harmonic profile to the idealized simulation environment. The RMS tracking error (E_{rms}) is defined as the ratio of the RMS value of the error signal to the RMS value of the reference voltage, as given in (10), and is calculated as a percentage.

$$E_{rms}(\%) = \sqrt{\frac{\sum_{k=1}^{N}\left[v_{ref}(k) - v_{out}(k)\right]^2}{\sum_{k=1}^{N}\left[v_{ref}(k)\right]^2}} \times 100. \qquad (10)$$

Here, $v_{ref}(k)$ and $v_{out}(k)$ represent the reference and output voltages at the kth sample, respectively, while N denotes the number of samples in the calculation window. E_{rms} was calculated over the last fundamental period after the system reached steady state.

Under linear load, the THD of the output voltage, which is 0.41% in open-loop operation is reduced to 0.25% with HCA method by controlling the fundamental component and the odd harmonics up to the 7th order. Under the same conditions, E_{rms} was 5.54% in open-loop operation, whereas it was 0.37% with HCA. Accordingly, HCA achieved an improvement of 39.0% in THD and 93.3% in E_{rms} compared to open-loop operation. The results obtained with open-loop operation and with the HCA method are presented in Fig. 4 and Fig. 5, respectively. As shown in Fig. 5(a), the output

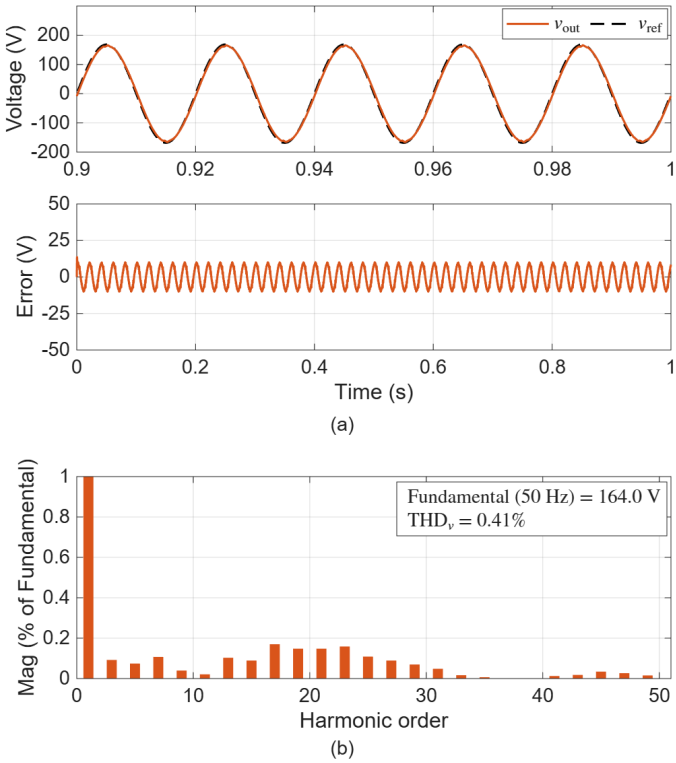


Fig. 4. Simulation results for open-loop operation under linear load. (a) Reference voltage, output voltage, and tracking error. (b) Harmonic spectrum of the output voltage.

voltage accurately tracks the reference voltage with HCA. In Fig. 5(b), it can be seen that all harmonic components of the error signal converge to zero within approximately 0.5 s. From the harmonic spectra in Fig. 4(b) and Fig. 5(c), it is observed that the harmonic components, which are already at low levels during open-loop operation, are further reduced with HCA method.

Under nonlinear load, the THD value of the output voltage and the E_{rms} value were obtained as 7.55% and 7.77% in open-loop operation. When odd harmonics up to the 21st order, along with the fundamental component, were controlled using the HCA method, the THD and E_{rms} values reduced to 0.43% and 0.51%, respectively. These results correspond to an improvement of 94.3% in THD and 93.4% in E_{rms} compared to open-loop operation. The results obtained from open-loop operation and the HCA method are illustrated in Fig. 6 and Fig. 7. While significant distortion is observed in the peak regions of the output voltage during open-loop operation in Fig. 6(a), Fig. 7(a) demonstrates that the output voltage maintains its sinusoidal waveform and tracks the reference voltage precisely with the proposed HCA method. In Fig. 7(b), it can be seen that all harmonic components of the error signal converge to zero within approximately 0.5 s. The harmonic spectra in Fig. 6(b) and Fig. 7(c) indicate that the harmonic components arising during open-loop operation are significantly suppressed by the HCA method.

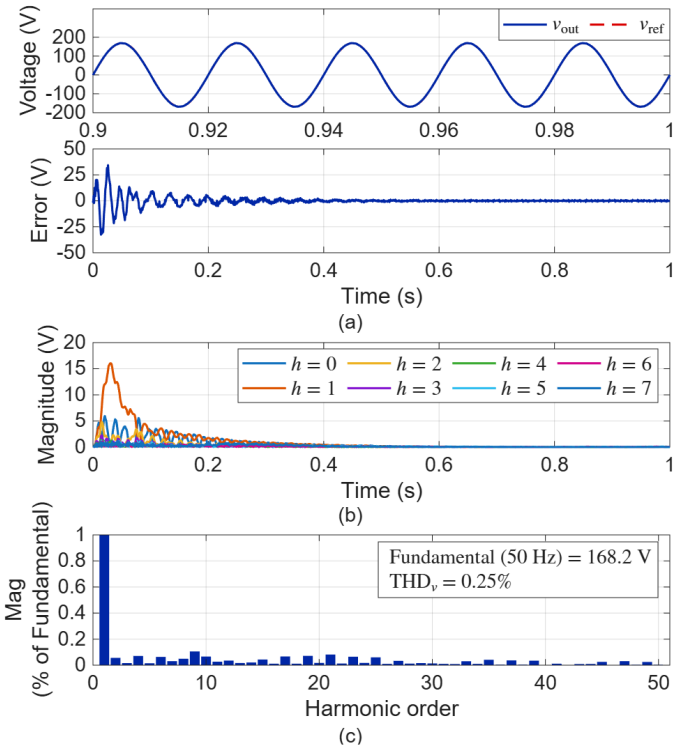


Fig. 5. Simulation results for the HCA method under linear load. (a) Reference voltage, output voltage, and tracking error. (b) Dispersion of error. (c) Harmonic spectrum of the output voltage.

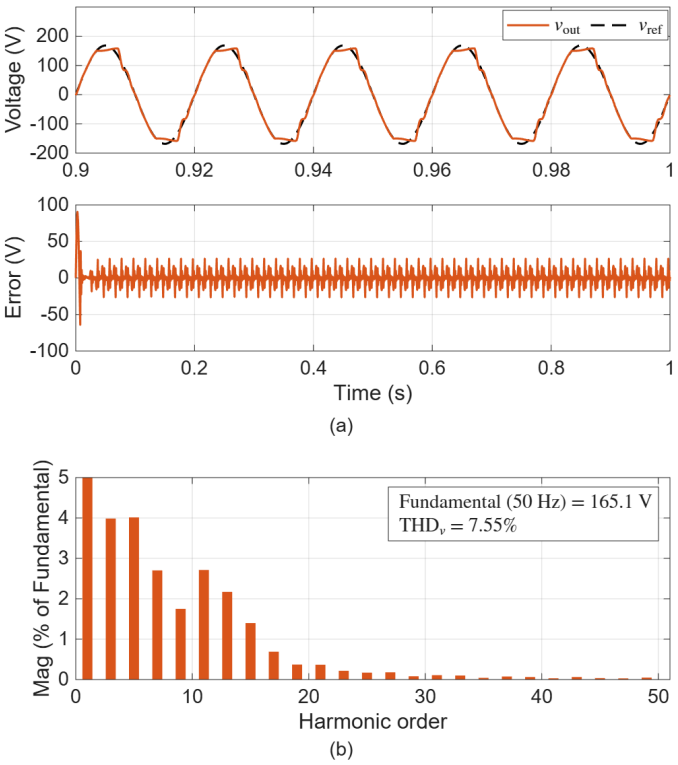


Fig. 6. Simulation results for open-loop operation under nonlinear load. (a) Reference voltage, output voltage, and tracking error. (b) Harmonic spectrum of the output voltage.

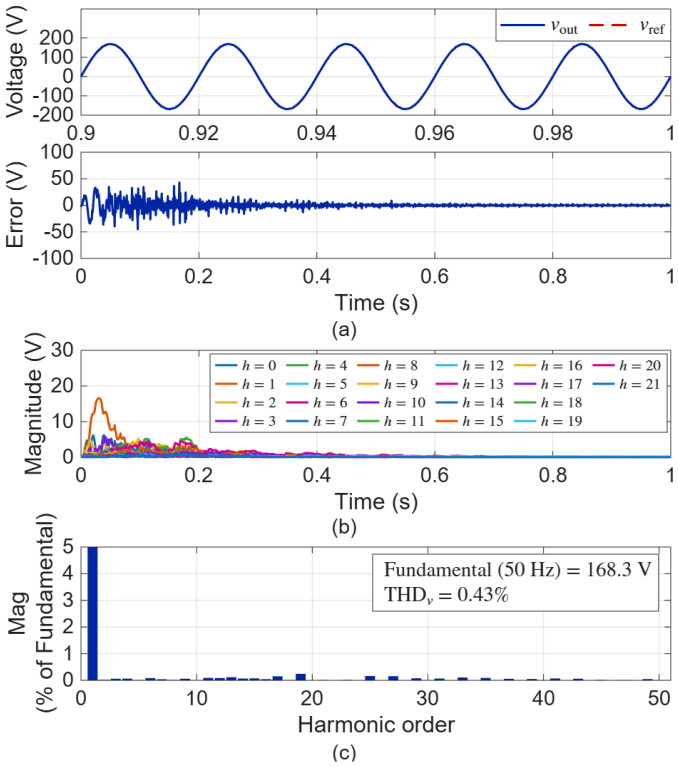


Fig. 7. Simulation results for the HCA method under nonlinear load. (a) Reference voltage, output voltage, and tracking error. (b) Dispersion of error. (c) Harmonic spectrum of the output voltage.

B. Experimental Studies

To verify the practical implementation of the proposed HCA method, a single-phase full-bridge inverter experimental setup was constructed. The experimental setup consists of a DC-bus power supply, a single-phase full-bridge inverter, an LC-type filter, linear and non-linear loads, an isolation circuit, and a voltage sensor circuit. The experimental setup developed for the nonlinear load is shown in Fig. 8. In the experiments conducted with the linear load, only the load was changed within the same setup and the linear load, the circuit structure of which is shown in Fig. 3 was connected instead of the nonlinear load. The system parameters used in the simulation studies and given in Table I were kept as the same in the experiments, and the inverter was supplied with a 200 V DC bus voltage. A Hall-effect voltage sensor (LEM LV25-P) was used to measure the load voltage. The control algorithm was implemented on the NUCLEO-G474RE development board from STMicroelectronics. Voltage waveforms were recorded using a Tektronix DPO3034 digital phosphor oscilloscope via a PINTEK DP-25 differential probe, while THD measurements were performed using a PCE-830 power and harmonic analyzer.

Under linear load, the output voltage THD value was measured as 2.1% in open-loop operation. When the fundamental component and odd harmonics up to the 7th order were controlled using the HCA method with the gain values employed in the simulation studies, the THD value decreased to 0.3%, achieving an 85.7% improvement compared to the open-loop operation. The experimental results obtained for open-loop

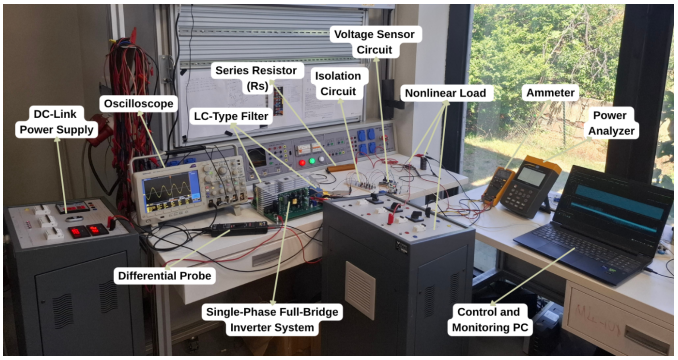
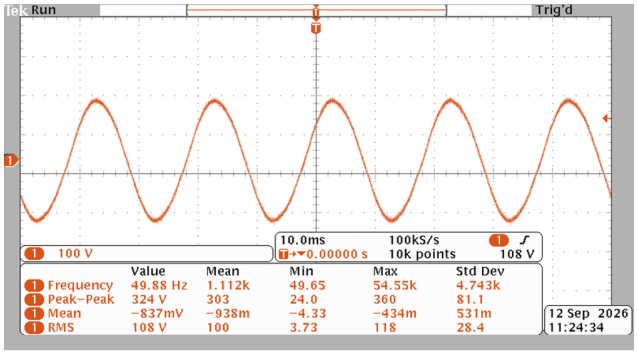


Fig. 8. Experimental setup constructed for the nonlinear load.



(a)

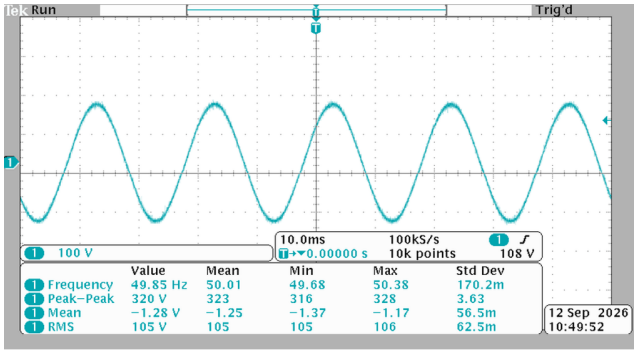


(b)

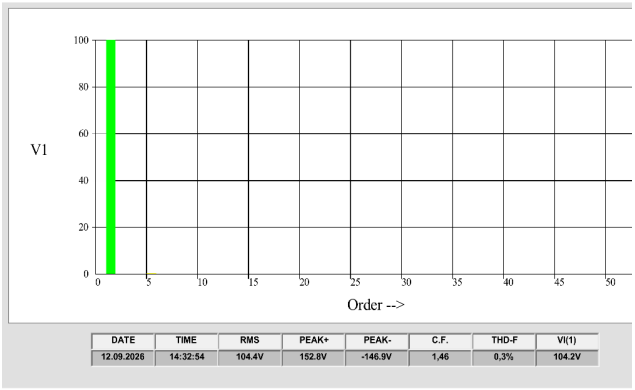
Fig. 9. Experimental results for open-loop operation under linear load. (a) Output voltage. (b) Harmonic spectrum of the output voltage.

operation and the HCA method are presented in Fig. 9 and Fig. 10, respectively. As shown in Fig. 9(a), in open-loop operation, the output voltage is close to sinusoidal, although there is slight distortion in the waveform. When the HCA method was applied, however, this distortion was eliminated and a smoother sinusoidal output voltage was obtained, as illustrated in Fig. 10(a). As can be seen from Fig. 9(b) and Fig. 10(b), the prominent 3rd harmonic and the low-level 7th harmonic observed during open-loop operation are effectively suppressed through the implementation of the HCA method.

Under non-linear load, the output voltage THD was measured as 6.8% in open-loop operation. When the fundamental component and odd harmonics up to the 21st order were controlled using the HCA method, the THD value reduced to



(a)



(b)

Fig. 10. Experimental results for the HCA method under linear load. (a) Output voltage. (b) Harmonic spectrum of the output voltage.

0.6%, achieving an improvement of 91.2% compared to open-loop operation. The experimental results obtained for open-loop operation and the HCA method are presented in Fig. 11 and Fig. 12, respectively. As shown in Fig. 11(a), significant distortion occurs in the peak regions of the output voltage due to the effect of the nonlinear load during open-loop operation. When the HCA method is applied, however, this distortion was significantly reduced and the output voltage achieved a clean sinusoidal output voltage waveform, as observed in Fig. 12(a). Comparing the harmonic spectra in Fig. 11(b) and Fig. 12(b), it is observed that the low-order odd harmonics, which are prominent during open-loop operation, are significantly suppressed by the HCA method.

The results obtained from the simulation and experimental studies are summarized in Table III. Under linear load conditions, the experimental open-loop THD value (2.1%) is significantly higher than that obtained in the simulation (0.41%). The primary reason for this difference is that dead time and switch conduction voltage drops were not taken into account in the simulation model [24], [35]. In contrast, when the HCA method was applied, the THD value decreased to 0.25% in simulation and 0.3% in experiment, and both studies yielded similar results. This indicates that the HCA method can also effectively suppress harmonics resulting from such non-ideal effects that are not included in the model. For nonlinear load conditions, close open-loop THD values were obtained in simulation (7.55%) and experimental measurement (6.8%). With the implementation of the HCA method, these

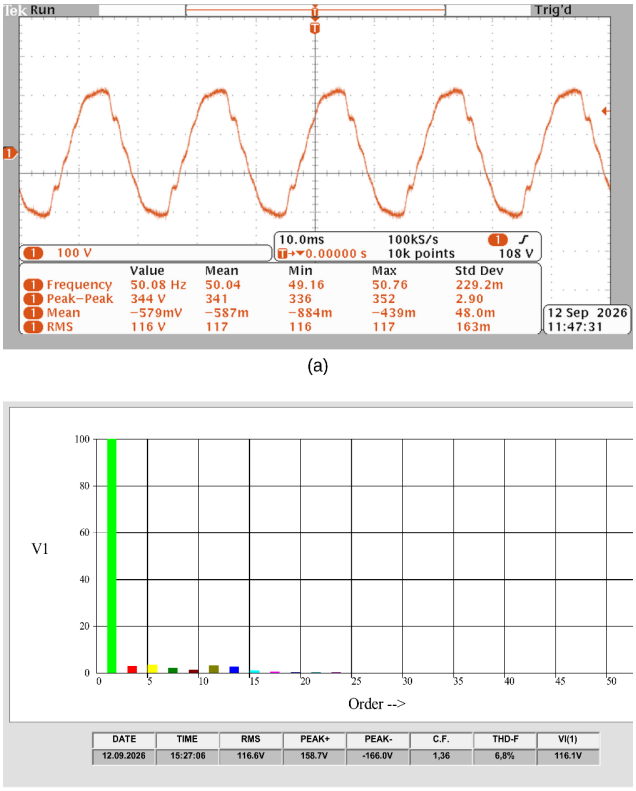


Fig. 11. Experimental results for open-loop operation under nonlinear load. (a) Output voltage. (b) Harmonic spectrum of the output voltage.

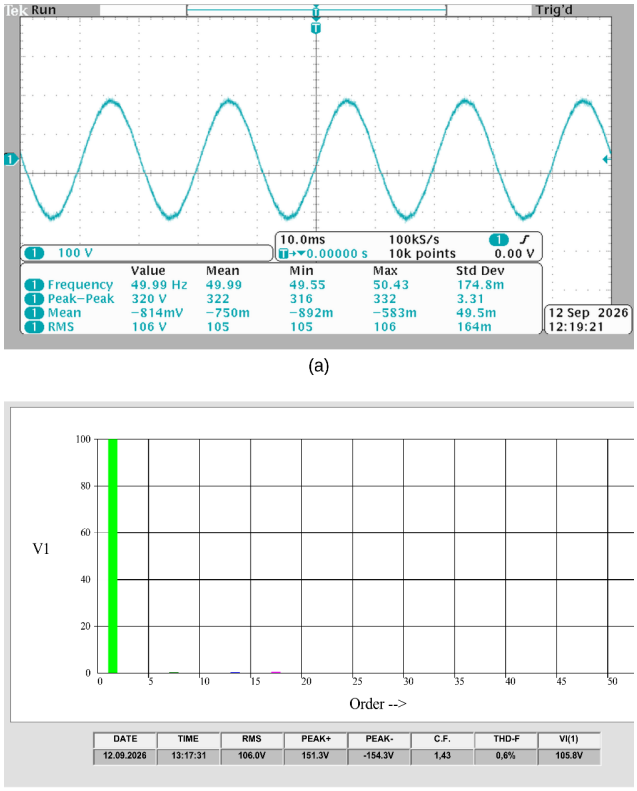


Fig. 12. Experimental results for the HCA method under nonlinear load. (a) Output voltage. (b) Harmonic spectrum of the output voltage.

TABLE III
COMPARISON OF SIMULATION AND EXPERIMENTAL RESULTS

Load type	Controlled harmonics	Operating mode	Simulation results		Experimental results	
			THD (%)	E_{rms} (%)	THD (%)	E_{rms} [a] (%)
Linear load	Fundamental and odd harmonics up to the 7th	Open-Loop Operation	0.41	5.54	2.1	-
		HCA	0.25	0.37	0.3	-
		Improvement (%)	39.0	93.3	85.7	-
Nonlinear load	Fundamental and odd harmonics up to the 21st	Open-Loop Operation	7.55	7.77	6.8	-
		HCA	0.43	0.51	0.6	-
		Improvement (%)	94.3	93.4	91.2	-

[a] *E_{rms} was not measured in the experimental studies; the reference tracking performance was evaluated using the oscilloscope recordings. The total harmonic distortion values were calculated considering harmonic components up to the 50th order.*

values were reduced to 0.43% and 0.6%, respectively. The fact that the simulation and experimental results are consistent with each other under both load conditions confirms the practical applicability of the proposed method.

VI. CONCLUSION

In this study, the Harmonic Control Array (HCA) method was applied to control the output voltage of an LC-filtered single-phase full-bridge inverter under linear and non-linear load conditions. The method disperses the error signal into its harmonic components, controlling each harmonic with independent gains. In this way, by selecting the harmonics to be controlled based on load conditions, both the tracking of the reference voltage and the suppression of the specified harmonics are achieved within a single structure. The proposed method was applied up to the 7th harmonic under linear load conditions and up to the 21st harmonic under nonlinear load conditions, and its performance was validated through both simulation and experimental studies. Throughout all simulation and experimental tests, the same gain values were utilized, and only the number of controlled harmonics was varied according to the load condition. In simulation studies, the output voltage THD was reduced from 0.41% to 0.25% under linear load and from 7.55% to 0.43% under nonlinear load. Furthermore, the E_{rms} error was decreased by more than 93% in both load scenarios, ensuring accurate tracking of the reference voltage. In experimental studies, the THD value was reduced from 2.1% to 0.3% under linear load conditions and from 6.8% to 0.6% under nonlinear load conditions. Additionally, a sinusoidal output voltage with low harmonic distortion was obtained for both load scenarios. Although power electronic non-idealities such as dead-time effects and switch conduction voltage drops were not included in the simulation model, the high consistency between the simulation and experimental outcomes strongly validates the practical applicability of the method. These results demonstrate that the proposed HCA framework delivers highly effective control performance, achieving precise reference tracking alongside effective harmonic suppression in real-world environments.

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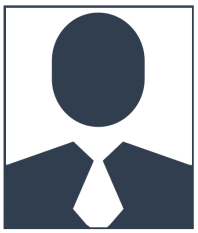
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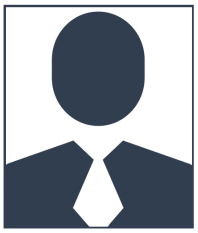
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